

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Founding engineer building full-stack features (Next.js + Firebase/GCP) for a provider-facing **SMART on FHIR** clinical decision support app that surfaces covered, guideline-backed medication alternatives inside the EHR prescribing workflow to prevent prior authorization.
- Extending a **Firebase Cloud Functions** backend on **Google Cloud** that builds alternatives from **RxNorm** normalization, formulary/prior-auth logic, and PubMed/guideline evidence, with **Gemini** constrained to synthesis while the backend deterministically controls eligibility and coverage.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's text-analysis platform (Python, **FastAPI**, sentence-transformers, React), productionizing a published NLP method into a self-serve web application for researchers.
- Designed versioned data infrastructure, a YAML model registry as single source of truth, and a structured data-quality warning system; grew the backend suite to **40 hermetic tests** running in CI without ML dependencies.

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, Express, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** through Mayden iaptus REST integrations; contributed to **\$340K+** in revenue, shipping in an Agile/Scrum team.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack: implemented **AWS KMS** encryption for clinical PII, hardened authentication and rate limiting, and remediated **20+ VAPT findings** with regression-tested fixes.

Projects

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a Spark/PySpark ETL pipeline over a **1.4B-row** NYC TLC dataset, cutting end-to-end runtime **~25%** (17–19 min to 14.3 min) via broadcast joins, Adaptive Query Execution, skew-join handling, shuffle-partition tuning, and column pruning; localized bottlenecks with Spark UI stage timings.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant with durable multi-step **workflow orchestration** (WorkflowEntrypoint + Durable Objects) so agent tasks survive process restarts and tool failures, persisting state in D1 and streaming responses to the client over WebSocket.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests** and provider routing/fallback.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, Java, C++, SQL, MongoDB, PostgreSQL, SQLite

Frameworks / Tools: Node.js, Express, React, Next.js, FastAPI, REST APIs, Docker, Git, GitHub Actions

Big Data & Systems: Spark, PySpark, Hadoop, Kafka, workflow orchestration, distributed systems, Linux profiling

ML & AI: PyTorch, Hugging Face Transformers, sentence-transformers, LLM agents, RAG, evaluation

Cloud & Platforms: GCP, AWS, Azure, Kubernetes, MongoDB Atlas, Linux

Open Source & Community

- Built **Underdogs Fitness** (Website), a production MERN platform with Stripe payments; live **3+ years**. - Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.